IoT Enabled Smart Helmet Breathalyzer

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***Abstract*—This paper is about designing and developing a smart helmet incorporating several modalities of sensing in order to maximize rider safety and situational awareness. The system utilizes an MQ3 sensor for real-time measurement of alcohol levels, an MPU6050 module to monitor head movement and looming impacts or falls, and a geofencing module to memorize pre-loaded regions of safety. The sensor readings are handled by a master microcontroller, which executes intelligent programs for learning risk factors and generating timely alarms. Test findings vindicate the efficacy of the integrated system for pre-crash hazard detection and positioning tracking with strong potential for preventing accident-related injuries. The suggested smart helmet is a suitable platform for the creation of wearable protection technology and future intelligent transportation systems.**

***Index Terms*—Smart Helmet, MQ3 Sensor, Alcohol Detection, MPU6050, Geofencing, Impact Detection, Microcontroller-Based Sensor Integration**

I. Introduction

The recent past has seen wearable technology revolutionize personal safety and prevention of accidents. In this paper, a smart helmet that aids in enhancing rider safety through the use of sophisticated sensing systems is discussed. The helmet has an MQ3 sensor that identifies alcohol in real time, thereby bringing an end to the risks associated with drunk driving. In addition, MPU6050 inertial measurement unit is employed for head movement tracking and for sensing potential impacts to enable early warning of accident conditions. Besides these sensors, there is a geofencing technique that enforces programmed safety boundaries for restricting the riders within the safe travel boundaries. Integrating information from these different sensors, the helmet can give real-time interventions and warnings to prevent accidents. This dual design approach not only overcomes the issues of current safety, but also contributes to the construction of the future intelligent transportation system as a fresh starting point for motorcyclist wearable safety technology and other utilization. The design achieves the best rider protection and functional reliability.

II. Literature Survey

Drastic increase in personal safety research from wearable technology has reared over the last three years with a special interest in smart helmets. Initial research focused on reintegration of basic sensor systems to track rider responses and sense impacts.

Most relevant work conducted here has been through the use of environmental and physiological sensors. There have been multiple architectures using alcohol sensors, say MQ3 sensor, that have been discussed through large sets of automobile safety systems for intoxication level measurement. Test results affirm onboard alcohol sensing as an effective avoidance measure by alerting the user and control system to risk onset before the accident.

Meanwhile, the MPU6050 sensor was of great interest as it is capable of measuring acceleration and angular velocity concurrently and is designed to measure fall and impact. Researchers pointed out that combined combined data of inertial sensors from sensor feedback is made possible for accurate tracking of head dynamics which is required in order to estimate the impact amplitude along with efficient making of emergency calls.

In addition to this, geofencing technology has also been touted for future use in future smart safety systems. Its use in smart transport systems for movement monitoring and spatial fence definition has proved to fence cars and riders in safe, pre-described regions. This blended approach, which uses the MQ3 sensor, MPU6050, and geofencing, is expected to provide real-time monitoring and warning, thus greatly improving the rider's safety.

III. System Design and Components

A. ESP32 Microcontroller

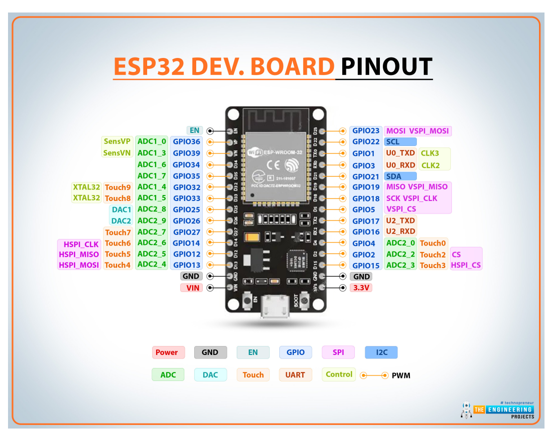

Fig. 1. ESP32-Microcontroller-Pin-Diagram

The ESP32 microcontroller is the processing core of the smart helmet. It is an efficient platform for sensor integration as well as wireless communication. The ESP32's renowned dual-core processor is fueled by massive peripheral support and therefore will work effectively in handling the data from a large number of sensors at a time. The built-in Wi-Fi feature provides a means of real-time sensor data transmission to external devices, such as smartphones or a central server without the need for any extra communication modules. The microcontroller supports Bluetooth connectivity, which is used for short-range communication or configuration. It is also appropriate for its low power design and energy-efficient operation for wearable applications with long battery life. This makes prototyping fast and easy across the Arduino IDE, and the ease of availability of libraries makes the interfacing to sensors such as MQ-3, MPU6050, and Neo6M GPS module very easy. Further, the number of I/O pins, ADC channels of the microcontroller, and the easily available communication interfaces (SPI, I2C, UART), provide good connectivity to sensors and good data acquisition. Thus, the ESP32 acts as the controller of smart helmets by processing the data and making intelligent decisions based on sensor inputs and then delivering any vital information like alcohol, impact accidents, or location, in real-time to the user.

B. MPU6050 Sensor

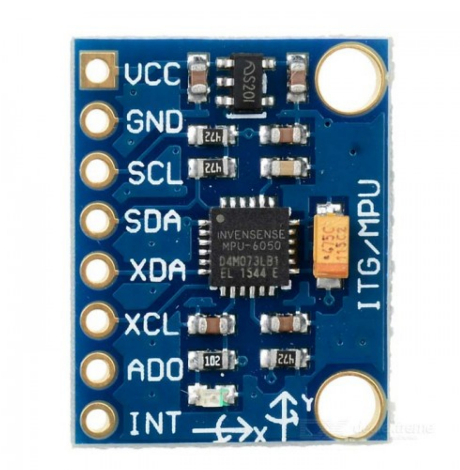

Fig. 2. MPU6050-Gyro-Sensor-Module

The MPU6050 sensor that consists of a 3-axis gyroscope and a 3-axis accelerometer and plays a important role in detecting crashes and falls on the smart helmet. It can sense acceleration and rotation movement to provides holistic data for the system to identify sudden impacts or unusual movements which indicates a crash or fall. The two-sensor configuration is optimal for detecting normal movement compared to emergency situations; rapid deceleration and rapid angular rotation, for example, could be used as an emergency alarm condition. MPU6050 only talks I2C protocol and, therefore, can easily be interfaced with the ESP32 microcontroller for processing the data read from the sensor in real-time. Advanced software can be deployed on the ESP32 to sense patterns of movement, filter out noise, and accurately identify a crash or a fall. The high sensitivity and response time of the sensor also mean that even low-level accident can be felt quickly, making the helmet more secure. Its small size and low power requirement also complement the wearable aspect of the project, and it is a critical piece in maintaining rider safety through enabling instant emergency response upon detection of critical motion occurrences.

C. NEO6M Sensor

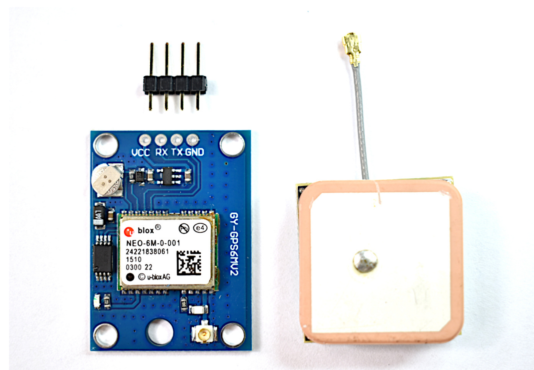

Fig. 3. NEO6M-Sensor-Module

The Neo6M GPS module is a key component of the smart helmet's geofencing and location tracking system. This module uses positioning data from multiple satellites to fix the geographic location of the helmet in real time. The Neo6M helps determine accurate latitude, longitude, and altitude information, which can be then applied to navigation as well as safety purposes. With the addition of the GPS module, the project can provide real-time monitoring, route tracking, and geofencing, a virtual perimeter that triggers an alert when the helmet passes defined boundaries. Data from the Neo6M is transmitted via serial interface to the ESP32 microcontroller, processed there, and transmitted over the built-in Wi-Fi for distant observation. The module is small and consumes low power, making it perfect for use for our smart helmet. Moreover, the fact that the module can host multiple GPS tracking software and APIs allows for wide customization so that it can be easily incorporated into larger safety and tracking systems by combining the location data of the helmet.The Neo6M GPS module enhances situational awareness and ensures that exact location data is always available to the rider and emergency responders.

D. 5V Battery

Fig. 4. 5V-Battery

A stable power supply is necessary for the smooth functioning of the smart helmet's sophisticated features and in this project, a 5V battery is used to fulfill this requirement. The 5V battery is selected for its compatibility with the ESP32 microcontroller and related sensors so that all modules can be provided with a constant voltage supply. This power supply drives the MQ3 sensor, MPU6050 sensor, and NeoM GPS module, and also maintains the uninterrupted working of the wireless communication system. The small size and weight of the battery make it a suitable option for wearable devices, where weight reduction and comfort is priority. Proper power distribution is facilitated through proper circuit design, which involves current limiting and power filtering to minimize noise and improve sensor performance. In brief, the 5V battery offers a stable, efficient, and portable power solution that supports the smart helmet's safety and monitoring features, providing reliable performance in different working conditions.

E. GEOFENCING

Fig. 5. Geofencing

Geofencing is one of the main features in the smart helmet to enhance rider safety and provide an added aspect of security. Using the precise location data provided by the Neo6M GPS module, geofencing allows virtual fences to be set up in a particular geographical area. When a vehicle moves in or out wearing the helmet from or into such predefined areas, the system provides indications in order to ensure that any unauthorized or sudden movement is reported immediately. The geofencing is put in place by establishing latitude and longitude coordinates to mark out an imaginary circle around key places or danger areas. ESP32 continuously processes the GPS readings and cross-checks them against geofenced boundaries. Anytime the helmet crosses these limits, the system evokes immediate reactions, such as messages on a connected phone or alerts to emergency centers. This application can also be particularly beneficial to monitor the movement of scooters and motorcycles around the city by parents or guardians for their child.

IV. Block Diagram

- Start: The helmet system starts, activating sensors like the MPU6050 (movement and fall detection), Neo6m (GPS tracking), and ESP32 (communication and processing).
- GPS Tracking: The Neo6m GPS tracker is always running to monitor the helmet location, with geofencing and live position tracking.

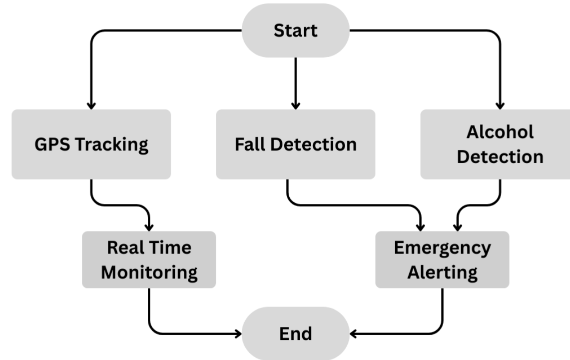

Fig. 6. Blockdiagram of IoT Enabled Smart Helmet Breathalyzer

- Fall Detection: The MPU6050 sensor identifies unusual movement patterns of a fall. If the system detects a fall, the emergency procedures are initiated.
- Alcohol Detection: The breathalyzer is mounted on the helmet and detects the alcohol content in the rider's breath. If detected at a set level of alcohol, further action like warning or limiting can be taken.
- Real-Time Monitoring: The system periodically retrieves and transmits location, movement, and alcohol status data for remote monitoring.
- Emergency Alerting: When the rider falls or the alcohol level is high, the ESP32 triggers an emergency alert through a network connection.
- End: The system ceases monitoring and alerting functions for safety compliance before shutdown or reset.

V. Implementation

A. Hardware Setup

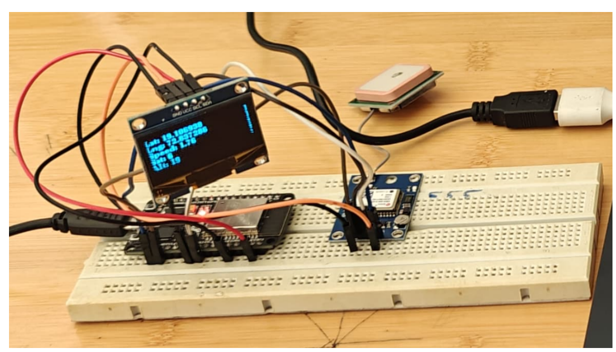

Fig. 7. Integration of Sensors with ESP32

- ESP32 Microcontroller: The ESP32 is the microcontroller that interfaces with all sensors, decodes the data, and communicates with other devices. It has in-built Wi-Fi and Bluetooth and is suitable for wireless communication and real-time monitoring.
- MQ3 Alcohol Sensor: MQ3 sensor is used for alcohol detection in the rider's breath. The sensor is sensitive and

beeps when a detected alcohol level crosses a specified level, thus avoiding drunk driving.

- MPU6050 Sensor: MPU6050 module is used for the head movement of the rider, angular speed, and acceleration. The module is utilized for the detection of abnormal movements of the rider, such as sudden impact or fall, which may signify an accident.
- Geofencing Module: The geofencing system uses a GPS module to track the position of the rider. Certain safe zones are stored in advance and alerted when traveling outside these areas. It also protects from accidents while traveling to no-go zones.
- Power Supply: The system is operated by a rechargeable lithium-ion battery that is adequate for a long time. Power management is used to ensure that each part operates safely for extended driving.

B. Software Development

- Sensor Data Acquisition: MQ3, MPU6050, and GPS data is read by the ESP32 microcontroller. Real-time data processing is performed through the ESP32 onboard ADC (for MQ3) and I2C communication (for MPU6050).
- Data Processing and Decision-Making: The algorithms process the sensor readings using software. For the MQ3 sensor, the software receives a reading of alcohol and compares it with a threshold level. The alarm is initiated once the threshold level is exceeded. For the MPU6050, the software senses sudden shocks or incorrect head movement, indicating the likelihood of an accident. The rider's position is tracked against pre-fed safety areas from GPS coordinates.
- Alert System: Whenever a threshold is crossed (e.g., alcohol level above normal, impact sensed, or exiting a safe area), the system sends alerts through audio and LED signals to the desired emergency contacts. In subsequent versions, wireless communication can be used to alert emergency contacts through a mobile app.
- Geofencing Algorithm: The GPS geofencing system tracks the rider's position and compares it with the set boundaries. A distance-based algorithm determines the rider's position relative to the boundary and sends alerts when approaching hazardous locations.
- Firmware: The firmware, implemented using the Arduino IDE, runs on the ESP32 microcontroller. It handles the sensor interface, data processing, and decision making and facilitates future updates or optimizations.

VI. Results and Discussion

The integrated system demonstrates effective real-time monitoring and prompt alert generation. Sensor data is processed and transmitted accurately to the IoT platform, where it is visualized and analyzed. The chatbot interface further simplifies user interaction by offering quick responses to common queries and troubleshooting steps.

VII. Conclusion

This paper presented a smart helmet system with numerous sensors for improved safety of the rider. Intoxication is identified by an MQ3 alcohol sensor, the MPU6050 shock and fall detection sensor identifies impacts, and GPS geofencing with GPS identifies locations to limit riders within safe areas. An ESP32 microcontroller controls the system with real-time analysis and feedback on warnings issued on time to avoid accidents.

The testing verified the effectiveness of the system to identify risks such as alcohol level, crashes, and geofence breaches with timely responses and high precision. However, power usage and GPS accuracy within urban environments were noted as issues, while the system was otherwise reliable.

In short, this intelligent helmet is an excellent step towards motorcyclist safety and can be improved in its future versions by adding wireless communication and better power management. It is a stepping stone to greater developments in wearable safety technologies.

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References

[1] V. Sudharsana, T. Govind Vineed, M. Mathews, S. Surendran, and M. Sabah, “Smart Helmet System Using Alcohol Detection for Vehicle Protection,” International Journal of Emerging Technology in Computer Science Electronics (IJETCSE), vol. 8, no. 1, Apr. 2014.

[2] N. Agarwal, A. K. Singh, and P. P. Singh, “Intelligent Helmet: Application of RF,” International Research Journal of Engineering and Technology (IRJET), vol. 2, no. 2, May 2015.

[3] A. Das, P. Das, and S. Goswami, “Brainwave and Alcohol Sensitising Helmet for Riders' Safety,” in Proceedings of the Eleventh IRF International Conference, 17 Aug. 2014.

[4] A. Das, P. Das, and S. Goswami, “Smart Helmet for Indian Bike Riders,” in Proceedings of the Eleventh IRF International Conference, 17 Aug. 2014.